



VIDOMxx-06

Aetos Domperidone Tablet 10mg

DESCRIPTION

Round, 6.0 mm in diameter, white to off-white coated tablet with shallow convex faces.

COMPOSITION

Each film-coated tablet contains Domperidone Maleate 12.73mg, equivalent to Domperidone 10 mg.

PHARMACODYNAMICS

Domperidone is a dopamine antagonist with anti-emetic properties.

Domperidone does not readily cross the blood-brain barrier. In domperidone users, especially adults, extrapyramidal side effects are very rare, but domperidone promotes the release of prolactin from the pituitary. Its anti-emetic effect may be due to a combination of peripheral (gastrokinetic) effects and antagonism of dopamine receptors in the chemoreceptor trigger zone, which lies outside the blood-brain barrier in the area postrema.

Studies in man have shown oral domperidone to increase lower oesophageal pressure, improve antroduodenal motility and accelerate gastric emptying.

There is no effect on gastric secretion.

PHARMACOKINETICS

Absorption

In fasting subjects, domperidone is rapidly absorbed after oral administration, with peak plasma concentrations at 30 to 60 minutes. The low absolute bioavailability of oral domperidone (approximately 15%) is due to an extensive first-pass metabolism in the gut wall and liver. Although domperidone's bioavailability is enhanced in normal subjects when taken after a meal, patients with gastro-intestinal complaints should take domperidone 15-30 minutes before a meal. Reduced gastric acidity impairs the absorption of domperidone. Oral bioavailability is decreased by prior concomitant administration of cimetidine and sodium bicarbonate. The time of peak absorption is slightly delayed and the AUC somewhat increased when domperidone is taken after a meal.

Distribution

Domperidone is 91 - 93% bound to plasma proteins.

Metabolism

Domperidone undergoes rapid and extensive hepatic metabolism by hydroxylation and N-dealkylation. In vitro metabolism experiments with diagnostic inhibitors revealed that CYP3A4 is a major form of cytochrome P-450 involved in the N-dealkylation of domperidone, whereas CYP3A4, CYP1A2 and CYP2E1 are involved in domperidone aromatic hydroxylation.

Excretion

Urinary and faecal excretions amount to 31 and 66% of the oral dose respectively. The proportion of the drug excreted unchanged is small (10% of faecal excretion and approximately 1% of urinary excretion). The plasma half-life after a single oral dose is 7 - 9 hours in healthy patients, but is prolonged in patients with severe renal insufficiency.

Special Populations

Hepatic Impairment

In patients with moderate hepatic impairment the unbound fraction is increased by 25%, and the terminal elimination half-life is prolonged from 15 to 23 hours. Patients with mild hepatic impairment have a somewhat lower systemic exposure, with no change in protein binding or terminal half-life.

Renal Impairment

In patients with severe renal insufficiency (serum creatinine > 6 mg/100 mL, i.e., > 0.6 mmol/L) the half-life of domperidone is increased from 7.4 to 20.8 hours, but plasma drug levels are lower than in patients with normal renal function. Very little unchanged drug (approximately 1%) is excreted via the kidneys.

INDICATIONS

Domperidone is indicated for the relief of the symptoms of nausea and vomiting.

This includes:

- Nausea and vomiting of functional, organic, infectious or dietary origin.

- Nausea and vomiting induced by:
 - Radiotherapy or drug therapy.
 - Dopamine agonists (such as L-dopa and bromocriptine) used in the treatment of Parkinson's disease.

CONTRAINDICATIONS

Aetos Domperidone is contraindicated in the following situations:

- Known hypersensitivity to domperidone or any of the excipients.
- Prolactin-releasing pituitary tumour (prolactinoma).
- In patients who have known existing prolongation of cardiac conduction intervals, particularly QTc, patients with significant electrolyte disturbances or underlying cardiac diseases such as congestive heart failure (see Warnings and Precautions).
- co-administration with QT-prolonging drugs
- co-administration with potent CYP3A4 inhibitors regardless of their QT prolonging effects (See Section Interactions).
- Whenever stimulation of gastric motility might be dangerous, e.g., in the presence of gastro-intestinal haemorrhage, mechanical obstruction or perforation.
- In patients with moderate or severe hepatic impairment

WARNINGS AND PRECAUTIONS

Cardiovascular effects

Domperidone has been associated with prolongation of the QT interval on the electrocardiogram. During post-marketing surveillance, there have been very rare cases of QT-prolongation and torsades de pointes in patients taking domperidone. These reports included patients with confounding risk factors, electrolyte abnormalities and concomitant treatment which may have been contributing factors (see Adverse Reactions).

Epidemiological studies showed that domperidone was associated with an increased risk of serious ventricular arrhythmias or sudden cardiac death (see Adverse Reactions). A higher risk was observed in patients older than 60 years, patients taking daily doses greater than 30 mg, and patients concurrently taking QTprolonging drugs or CYP3A4 inhibitors.

Domperidone should be used at the lowest effective dose in adults and children.

Domperidone is contraindicated in patients with known existing prolongation of cardiac conduction intervals, particularly QTc, in patients with significant electrolyte disturbances (hypokalaemia, hyperkalaemia, hypomagnesaemia), or bradycardia, or in patients with underlying cardiac diseases such as congestive heart failure due to increased risk of ventricular arrhythmia (see Contraindications).

Electrolyte disturbances (hypokalaemia, hyperkalaemia, hypomagnesaemia) or bradycardia are known to be conditions increasing the proarrhythmic risk.

Treatment with domperidone should be stopped if signs or symptoms occur that may be associated with cardiac arrhythmia, and the patients should consult their physician.

Patients should be advised to promptly report any cardiac symptoms.

Ability to drive and use machines: Dizziness and somnolence have been observed following use of domperidone (see Adverse Reactions). Therefore, patients should be advised not to drive or use machinery or engage in other activities requiring mental alertness and coordination until they have established how Domperidone affects them.

PREGNANCY AND LACTATION

Pregnancy

There are limited post-marketing data on the use of domperidone in pregnant women. A study in rats has shown reproductive toxicity at a high, maternally toxic dose. The potential risk for humans is unknown. Therefore, Domperidone should only be used during pregnancy when justified by the anticipated therapeutic benefit.

Breast-feeding

The amount of domperidone that could be ingested by an infant through breast milk is low. The maximal relative infant dose (%) is estimated to be about 0.1% of the maternal weight- adjusted dosage. It is not known whether this is harmful to the newborn. Therefore, breast-feeding is not recommended for mothers who are taking Domperidone.



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INTERACTIONS WITH OTHER MEDICAMENTS

The main metabolic pathway of domperidone is through CYP3A4. In vitro and human data show that the concomitant use of drugs that significantly inhibit this enzyme may result in increased plasma levels of domperidone. Co-administration of domperidone with potent CYP3A4 inhibitors which have been shown to cause QT interval prolongation is contraindicated (See Section Contraindications).

Caution should be exercised when domperidone is co-administered with potent CYP3A4 inhibitors which have not been shown to cause QT interval prolongation such as indinavir and patients should be monitored closely for signs or symptoms of adverse reactions.

Caution should be exercised when domperidone is co-administered with drugs which have been shown to cause QT interval prolongation and patients should be monitored closely for signs or symptoms of cardiovascular adverse reactions. Examples include:

- anti-arrhythmics class IA (e.g., disopyramide, quinidine)
- anti-arrhythmics class III (e.g., amiodarone, dofetilide, dronedarone, ibutilide, sotalol)
- certain antipsychotics (e.g., haloperidol, pimozide, sertindole)
- certain antidepressants (e.g., citalopram, escitalopram)
- certain antibiotics (e.g., levofloxacin, moxifloxacin)
- certain antifungal agents (e.g., pentamidine)
- certain antimalarial agents (e.g., halofantrine)
- certain gastro-intestinal drugs (e.g., dolasetron)
- certain drugs used in cancer (e.g., toremifene, vandetanib)
- certain other drugs (e.g., bepridil, methadone)

Antacids or antisecretory agents should not be taken simultaneously with oral formulations of Domperidone, as they lower the oral bioavailability of domperidone. When used concomitantly, Domperidone should be taken before meals and antacids or antisecretory agents after meals.

MAIN SIDE / ADVERSE REACTIONS

Immune System Disorders

Very rare: Anaphylactic Reaction (including Anaphylactic Shock)

Psychiatric Disorders

Very rare: Agitation, Nervousness

Nervous System Disorders

Very rare: Dizziness, Extrapryamidal Disorder, Convulsion

Cardiac Disorders

Very rare: Ventricular arrhythmias, QTc prolongation, Torsade de Pointes, Sudden cardiac death (see Warnings and Precautions)

Skin and Subcutaneous Tissue Disorders

Very rare: Angioedema

Renal and Urinary Disorders

Very rare: Urinary Retention

Investigations

Very rare: Liver Function Test Abnormal, Blood Prolactin Increased

Pediatric population

In postmarketing experience, there were no differences in the safety profile of adults and children.

OVERDOSE

Overdose has been reported primarily in infants and children. Symptoms of overdose may include agitation, altered consciousness, convulsion, disorientation, somnolence and extrapyramidal reactions.

Treatment:

There is no specific antidote to domperidone. Close medical supervision and supportive therapy is recommended. Anticholinergic or anti-Parkinson drugs may be helpful in controlling the extrapyramidal reactions. It is advisable to contact a poison control center to obtain the latest recommendations for the management of an overdose.

DOSAGE AND ADMINISTRATION

Oral.

It is recommended to take Aetos Domperidone 15 – 30 minutes before meals. If taken after meals, absorption of the drug is somewhat delayed.

Adults and adolescents ≥ 12 years of age and weighing ≥ 35 kg & children < 12 years of age and weighing ≥ 35 kg

The dose of Aetos Domperidone should be the lowest effective dose for the individual situation (typically 30 mg/day) and can be increased if necessary to a maximum daily oral dose of 40 mg. Usually, the maximum treatment duration should not exceed one week for the treatment of acute nausea and vomiting. If nausea and vomiting persists for longer than one week, patients should consult their physician. For other indications, the initial duration of treatment is up to four weeks. If treatment exceeds four weeks, patients should be re-evaluated and the need for continued treatment reassessed.

Formulation (domperidone per unit)	Dosage	Maximum dose per day
Tablets (10mg/tablet)	1 tablet three to four times per day	40 mg (4 x 10 mg tablet)
Oral suspension (1 mg/ml)	10 mL three to four times per day	40 mg (40 mL of 1 mg/mL oral suspension)

Adults and adolescents (≥ 12 years of age) weighing < 35 kg

The dose of Aetos Domperidone should be the lowest effective dose. The total daily dose is dependent on weight. Usually, the maximum treatment duration should not exceed one week for the treatment of acute nausea and vomiting. For other indications, the initial duration of treatment is up to four weeks. If treatment exceeds four weeks, patients should be reevaluated and the need for continued treatment reassessed. Due to the need for accurate dosing, tablets are unsuitable for use in adults and adolescents weighing less than 35 kg.

Formulation (domperidone per unit)	Dosage	Maximum dose per day
Oral suspension (1 mg/ml)	0.25 mg/kg three to four times per day	1 mg/kg but no more than 35 mL (35mg)

Infants and children < 12 years of age and weighing < 35 kg

The efficacy of Aetos Domperidone has not been established in infants and children < 12 years of age and weighing < 35 kg.

Renal impairment

Since the elimination half-life of domperidone is prolonged in severe renal impairment (serum creatinine > 6 mg/100 mL, i.e. > 0.6 mmol/L), the dosing frequency of Aetos Domperidone should be reduced to once or twice daily, depending on the severity of the impairment, and the dose may need to be reduced. Patients with severe renal impairment should be reviewed regularly.

Hepatic impairment

Aetos Domperidone is contraindicated for patients with moderate (Child-Pugh 7 to 9) or severe (Child-Pugh >9) hepatic impairment. Dose adjustment is not required for patients with mild (Child-Pugh 5 to 6) hepatic impairment.

Storage: Store below 30°C.

Presentation / Packing: Blister pack of 10 x 10's

Product Registration Holder: Abiotide Sdn. Bhd.
121, Jalan Tunku Abdul Rahman,
30010 Ipoh, Perak, Malaysia.

Manufactured by: HOVID Berhad
Lot 56442, 7 ½ Miles, Jalan Ipoh / Chemor,
31200 Chemor, Perak, Malaysia.

Revision date: September 2022